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Problem 1. *Here we use the SCF data set again. The second last variable ("V11") is Wealth ($1000), and in this assignment you will use this series only. First, obtain the sample median of Wealth (call it* $\widehat{med}(W)$*) we simply use the 51st smallest observation.*

(i) Use the nonparametric bootstrap to estimate the variance of the sample median ($var(\widehat{med}(W))$). *Try also the nonparametric bootstrap bias correction to* $\widehat{med}(W)$. *The number of bootstrap reps is 1000.*

(ii) (Out of the blue) you think that the wealth series is IID, lognormal, with unknown mean and variance parameters. Apply the parametric bootstrap to estimate $var(\widehat{med}(W))$. *The number of bootstrap reps is still 1000. Also use the asymptotic variance formula (the* $1/(4f^2)$ *thing, where you estimate* f *from the parametric MLE) to estimate the same thing.*

Solution. Implementation in Python:

```
import numpy as np
import pandas as pd

data = pd.read_csv('metdata.txt', sep=r'\s+')
wealth = data['V11'].values
n = len(wealth)

# Sample median (51st smallest)
med = np.sort(wealth)[50]

# (i) Nonparametric bootstrap
np.random.seed(123)
B = 1000
med_np = np.array([np.sort(np.random.choice(wealth, n,
                   replace=True))[50] for _ in range(B)])
var_np = np.var(med_np, ddof=1)
bias_np = np.mean(med_np) - med
med_bc = med - bias_np

# (ii) Parametric bootstrap (lognormal)
lw = np.log(wealth)
mu, sig = np.mean(lw), np.sqrt(np.var(lw, ddof=0))
med_para = np.array([np.sort(np.exp(np.random.normal(mu, sig, n)))[50]
                     for _ in range(B)])
var_para = np.var(med_para, ddof=1)

# Asymptotic variance: 1/(4nf^2) where f is density at median
f_med = 1 / (np.exp(mu) * sig * np.sqrt(2*np.pi))
var_asy = 1 / (n * 4 * f_med**2)
```

(i) For nonparametric bootstrap, resample $\{W_i^{*b}\}_{i=1}^n$ with replacement from $\{W_i\}_{i=1}^n$ for $b = 1, \ldots, 1000$, computing the median (51st smallest) each time.
Bootstrap variance: $\widehat{\text{var}}_{\text{np}} = 1.115$ with SE $= 1.056$. Bootstrap bias: $\widehat{\text{bias}} = 0.171$. Bias-corrected: $\widehat{\text{med}}_{\text{BC}} = 7.032$.

(ii) Under lognormal assumption $W_i = e^{X_i}$ where $X_i \sim N(\mu, \sigma^2)$, MLE gives $\hat{\mu} = 1.749$ and $\hat{\sigma} = 1.471$. For parametric bootstrap, draw $X_i^{*b} \sim N(\hat{\mu}, \hat{\sigma}^2)$, compute $W_i^{*b} = e^{X_i^{*b}}$, and take the median. Bootstrap variance: $\widehat{\text{var}}_{\text{para}} = 1.240$ with SE $= 1.114$. For asymptotic variance, the median is e^{μ} with density $f(e^{\mu}) = (e^{\mu}\sigma\sqrt{2\pi})^{-1}$, giving $\widehat{\text{var}}_{\text{asy}} = n^{-1} \cdot (4\hat{f}^2)^{-1} = 1.124$ with SE $= 1.060$.

□

Problem 2. *Suppose we observe a random sample* $\{(x_i, y_i)\}_{i=1}^n$ *of a bivariate random variable* (x, y)*, with* $E[(x, y)'] = (\mu_x, \mu_y)'$ *and* $\text{var}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{pmatrix}$*. Our parameter of interest is* $\theta = \max(\mu_x, \mu_y)$*. A natural estimator is* $\hat{\theta} = \max(\bar{x}, \bar{y})$ *where* $\bar{x} = \frac{1}{n}\sum_{i=1}^n x_i$ *and* $\bar{y} = \frac{1}{n}\sum_{i=1}^n y_i$*.*

(i) Obtain the asymptotic distribution of $\hat{\theta}$ *(of course, we need appropriate centering and scaling to obtain meaningful asymptotics). Note* $\mu_x = \mu_y$ *may or may not hold.*

(ii) Suppose we nonparametrically bootstrap $\hat{\theta}$ *to obtain* $\hat{\theta}^* = \max(\bar{x}^*, \bar{y}^*)$*. Here we simply resample the pairs* $\{(x_i, y_i)\}_{i=1}^n$ *according to its empirical distribution. Obtain the asymptotic distribution of* $\hat{\theta}^*$ *with appropriate centering and scaling. Discuss the consistency of the bootstrap.*

Solution. (i) By CLT,

$$\sqrt{n}\begin{pmatrix} \bar{x} - \mu_x \\ \bar{y} - \mu_y \end{pmatrix} \xrightarrow{d} \begin{pmatrix} Z_x \\ Z_y \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{pmatrix}\right)$$

where Z_x and Z_y are independent. The estimator is $\hat{\theta} = g(\bar{x}, \bar{y})$ where $g(a, b) = \max(a, b)$.

Case 1: $\mu_x > \mu_y$. Here $\theta_0 = \mu_x$ and $g(a, b) = a$ with gradient $\nabla g = (1, 0)'$. By Delta Method, $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} Z_x \sim N(0, \sigma_x^2)$. Case 2: $\mu_y > \mu_x$. Here $\theta_0 = \mu_y$ with gradient $\nabla g = (0, 1)'$, yielding $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} Z_y \sim N(0, \sigma_y^2)$. Case 3: $\mu_x = \mu_y = \mu$. Here $\theta_0 = \mu$ and g is not differentiable. By CMT, $\sqrt{n}(\hat{\theta} - \theta_0) = \max(\sqrt{n}(\bar{x} - \mu), \sqrt{n}(\bar{y} - \mu)) \xrightarrow{d} \max(Z_x, Z_y)$.

(ii) By bootstrap CLT,

$$\sqrt{n}\begin{pmatrix} \bar{x}^* - \bar{x} \\ \bar{y}^* - \bar{y} \end{pmatrix} \xrightarrow{d^*} \begin{pmatrix} Z_x^* \\ Z_y^* \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \hat{\Sigma}\right)$$

where

$$\hat{\Sigma} = \begin{pmatrix} \hat{\sigma}_x^2 & \hat{\sigma}_{xy} \\ \hat{\sigma}_{xy} & \hat{\sigma}_y^2 \end{pmatrix}$$

with $\hat{\sigma}_{xy} \xrightarrow{p} 0$ by LLN.

Case 1: $\mu_x > \mu_y$. Since $\bar{x} > \bar{y}$ w.p.a.1, $\sqrt{n}(\hat{\theta}^* - \hat{\theta}) = \sqrt{n}(\bar{x}^* - \bar{x}) \xrightarrow{d^*} Z_x^* \sim N(0, \hat{\sigma}_x^2)$. Since $\hat{\sigma}_x^2 \xrightarrow{p} \sigma_x^2$, the bootstrap is consistent. Case 2: $\mu_y > \mu_x$. Similarly, $\bar{y} > \bar{x}$ w.p.a.1 and $\sqrt{n}(\hat{\theta}^* - \hat{\theta}) \xrightarrow{d^*} Z_y^* \sim N(0, \hat{\sigma}_y^2) \xrightarrow{p} N(0, \sigma_y^2)$. The bootstrap is consistent. Case 3: $\mu_x = \mu_y = \mu$. The bootstrap is centered at $(\bar{x}, \bar{y})$. Since $\sqrt{n}(\bar{x} - \bar{y}) \xrightarrow{d} N(0, \sigma_x^2 + \sigma_y^2)$, $\bar{x} \neq \bar{y}$ with probability 1 for any finite n. The bootstrap limiting distribution depends on the sample: If the sample has $\bar{x} > \bar{y}$, the bootstrap (as in Case 1) yields $\sqrt{n}(\hat{\theta}^* - \hat{\theta}) \xrightarrow{d^*} N(0, \hat{\sigma}_x^2) \xrightarrow{p} N(0, \sigma_x^2)$. If the sample has $\bar{y} > \bar{x}$, the bootstrap (as in Case 2) yields $\sqrt{n}(\hat{\theta}^* - \hat{\theta}) \xrightarrow{d^*} N(0, \hat{\sigma}_y^2) \xrightarrow{p} N(0, \sigma_y^2)$. The true limiting distribution is $\max(Z_x, Z_y)$. The bootstrap limiting distribution is random (either $N(0, \sigma_x^2)$ or $N(0, \sigma_y^2)$ w.p. $\approx 1/2$) and does not converge to the true distribution. The bootstrap is inconsistent. □

Problem 3. *Suppose we observe a random sample $\{x_i\}_{i=1}^n$ of a q-dimensional random vector x, with $E[x] = \mu = (\mu_1, ..., \mu_q)'$ and $Var(x) = \Sigma$. Define $[x]_+ = (\max(x_1, 0), ..., \max(x_q, 0))'$. Consider $J = n\bar{x}'W\bar{x}$ and $J_+ = n[\bar{x}]_+'W[\bar{x}]_+$ where W is a $q \times q$ fixed positive definite matrix.*

(i) Analyze the asymptotic behavior of J and J_+ in: (A) $\mu_1 = \mu_2 = ... = \mu_q > 0$ and (B) $\mu_1 = \mu_2 = ... = \mu_q = 0$.

(ii) Now resample $\{x_i\}_{i=1}^n$ and calculate $\bar{x}^* = \frac{1}{n}\sum_{i=1}^n x_i^*$. Define $J^a = n(\bar{x}^*)'W\bar{x}^*$, $J^b = n[\bar{x}^* - \bar{x}]'W[\bar{x}^* - \bar{x}]$, $J_+^a = n[\bar{x}^*]_+'W[\bar{x}^*]_+$, $J_+^b = n([\bar{x}^*]_+ - [\bar{x}]_+)'W([\bar{x}^*]_+ - [\bar{x}]_+)$. Analyze the asymptotic behavior of J^a, J^b, J_+^a and J_+^b in cases (A) and (B).

Solution. (i) By CLT, $\sqrt{n}(\bar{x} - \mu) \xrightarrow{d} Z \sim N(0, \Sigma)$. By bootstrap CLT, $\sqrt{n}(\bar{x}^* - \bar{x}) \xrightarrow{d^*} Z^* \sim N(0, \hat{\Sigma})$ where $\hat{\Sigma} \xrightarrow{p} \Sigma$. Let $Z_n = \sqrt{n}(\bar{x} - \mu)$ and $Z_n^* = \sqrt{n}(\bar{x}^* - \bar{x})$. Case A: $\mu_1 = \cdots = \mu_q = \mu > 0$. For $J = n\bar{x}'W\bar{x}$, let $g(\mu) = \mu'W\mu$ with $\nabla g(\mu) = 2W\mu$. By Delta Method, $\sqrt{n}(\bar{x}'W\bar{x} - \mu'W\mu) \xrightarrow{d} N(0, 4\mu'W\Sigma W\mu)$. For $J_+ = n[\bar{x}]_+'W[\bar{x}]_+$, since $\bar{x} \xrightarrow{p} \mu > 0$, have $[\bar{x}]_+ = \bar{x}$ w.p.a.1, so J_+ has the same asymptotic distribution as J. Case B: $\mu = 0$. For $J = n\bar{x}'W\bar{x} = (\sqrt{n}\bar{x})'W(\sqrt{n}\bar{x})$, by CMT, $J \xrightarrow{d} Z'WZ$. For $J_+ = n[\bar{x}]_+'W[\bar{x}]_+ = ([\sqrt{n}\bar{x}]_+)'W([\sqrt{n}\bar{x}]_+)$, by CMT, $J_+ \xrightarrow{d} [Z]_+'W[Z]_+$.

(ii) Case A: $\mu > 0$. For J^a, by bootstrap Delta method, $\sqrt{n}((\bar{x}^*)'W(\bar{x}^*) - \bar{x}'W\bar{x}) \xrightarrow{d^*} N(0, 4\bar{x}'W\hat{\Sigma}W\bar{x}) \xrightarrow{p} N(0, 4\mu'W\Sigma W\mu)$. The bootstrap is consistent. For $J^b = (Z_n^*)'W(Z_n^*) \xrightarrow{d^*} Z'WZ$, which does not match the normal distribution. The bootstrap is inconsistent. For J_+^a, since $\bar{x} \xrightarrow{p} \mu > 0$, have $J_+^a = J^a$ asymptotically. The bootstrap is consistent. For J_+^b, since all components are positive w.p.a.1, $J_+^b = J^b$ asymptotically. The bootstrap is inconsistent. Case B: $\mu = 0$. For J^a, write $J^a = (\sqrt{n}\bar{x}^*)'W(\sqrt{n}\bar{x}^*) = (Z_n + Z_n^*)'W(Z_n + Z_n^*)$. In the joint limit, $J^a \xrightarrow{d} (Z + Z^*)'W(Z + Z^*)$ where Z and Z^* are independent $N(0, \Sigma)$. This does not match $Z'WZ$. The bootstrap is inconsistent. For $J^b = (Z_n^*)'W(Z_n^*) \xrightarrow{d^*} Z'WZ$. The bootstrap is consistent. For $J_+^a = ([Z_n + Z_n^*]_+)'W([Z_n + Z_n^*]_+) \xrightarrow{d} ([Z + Z^*]_+)'W([Z + Z^*]_+)$, which does not match $[Z]_+'W[Z]_+$. The bootstrap is inconsistent. For J_+^b, the positive part function $[\cdot]_+$ is not differentiable at zero. Since $\bar{x} = O_p(n^{-1/2})$, the elements are local to the kink point. Recentering fails because $[\bar{x}^*]_+ - [\bar{x}]_+ \neq [\bar{x}^* - \bar{x}]_+$. The bootstrap is inconsistent.

In Case A, non-recentered versions (J^a, J_+^a) are consistent while recentered versions (J^b, J_+^b) fail. In Case B, the recentered version J^b is consistent while non-recentered J^a fails, but recentering for J_+ remains inconsistent due to non-differentiability at the boundary. □